

Omar Essam El-Din Mohamed

Data Engineering & Machine Learning Lead

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Credly

PROFESSIONAL SUMMARY

An accomplished Data Engineering and Machine Learning Lead with 6+ years of specialized experience in AI development, cloud architecture, and team leadership. Expert in building scalable data pipelines across AWS, Azure, and GCP, implementing advanced ML/AI solutions including RAG systems, deep learning models, and agentic AI. Proven track record of leading cross-functional teams using Agile methodologies, delivering enterprise-grade solutions, and driving digital transformation initiatives. Combines strong technical expertise with strategic project management to deliver innovative solutions that optimize business operations and enhance decision-making capabilities.

CORE COMPETENCIES

Cloud & Infrastructure

- AWS (EC2, S3, Lambda, SageMaker, Redshift)
- Azure (ML, Data Factory, Synapse, AKS)
- GCP (BigQuery, Vertex AI, TPUs, GKE)
- Kubernetes, Docker, Terraform

Machine Learning & AI

- Deep Learning (PyTorch, TensorFlow)
- LLMs, RAG, Vector Databases
- MLOps, Model Deployment
- Computer Vision, NLP

Data Engineering & Analytics

- ETL/ELT Pipelines, Apache Airflow
- Spark, Hadoop, Databricks, Snowflake
- SQL, Python, R, Java
- Power BI, Tableau, Advanced Analytics

Leadership & Project Management

- Agile/Scrum/Lean Methodologies
- Team Leadership, Stakeholder Management
- Jira, GitLab CI/CD, Jenkins
- Risk Management, Strategic Planning

PROFESSIONAL EXPERIENCE

Data Engineering & Machine Learning Leadership

Data Engineering And Machine Learning Lead — Rowad Modern Engineering Jan 2025 – Present
Leading ML/AI initiatives and advanced analytics solutions

- Architected and deployed production-ready RAG chatbot using vector databases (Pinecone, Chroma) and LangChain, improving internal knowledge access by 60%
- Built comprehensive full-stack analytics platform integrating Oracle ERP with on-premises data warehouse, enabling real-time business intelligence and decision-making

- Implemented advanced deep learning models using PyTorch and TensorFlow, including CNNs for image classification and transformer architectures for NLP tasks
- Led cross-functional AI team of 8 engineers using Agile/Scrum methodologies, achieving 95% sprint completion rate and reducing development cycles by 40%
- Leveraged NVIDIA 4090 GPUs for high-performance AI agent development, automating coding workflows and reducing manual debugging time by 70%
- Deployed agentic AI systems with goal-based and utility-based agents for autonomous task execution and workflow optimization
- Implemented MLOps practices using Kubernetes, MLflow, and CI/CD pipelines (GitLab CI, Jenkins) for automated model deployment and monitoring
- Mentored junior engineers in advanced ML techniques, cloud technologies, and best practices in data engineering and ML

Data Engineering Lead — Rowad Modern Engineering

Nov 2022 – Dec 2024

Scaling data infrastructure and leading engineering teams

- Established and led data engineering team of 12 professionals, implementing Agile practices with Scrum for project tracking and sprint management
- Designed and deployed enterprise-scale data pipelines across AWS (S3, Redshift, Lambda), Azure (Data Factory, Synapse), and GCP (BigQuery, Dataflow)
- Built real-time streaming data solutions using Apache Kafka and Spark, processing 10M+ records daily with 99.9% uptime
- Architected cloud-native data warehouse solutions using Snowflake and Databricks, reducing query response times by 75%
- Led stakeholder management and risk mitigation strategies, ensuring project delivery within budget and timeline constraints

Data Scientist & Scrum Master — Rowad Modern Engineering

May 2022 – Oct 2022

Advanced analytics, predictive modeling, and agile team facilitation

- Developed sophisticated ML models for cash flow predictions using ensemble methods (XGBoost, Random Forest) and time series analysis
- Implemented feature engineering pipelines with automated data preprocessing, achieving 15% improvement in model accuracy
- Facilitated Scrum ceremonies and coached cross-functional team of 6 members in Agile methodologies, improving sprint velocity by 25%

Senior Data Engineer & DBA — Rowad Modern Engineering

Nov 2020 – Apr 2022

Infrastructure automation, optimization, and database administration

- Automated complex business processes using Python and SQL, saving 200+ hours monthly through intelligent workflow optimization
- Integrated data from multiple enterprise sources into centralized data warehouse including Oracle ERP, Oracle Unifier, CRM, Primavera P6 EPPM, and HR Hits systems
- Managed database administration including user management, configuration optimization, table design, and performance monitoring across production systems

Data Engineer — Rowad Modern Engineering

Jan 2020 – Oct 2020

Infrastructure setup and data pipeline development

- Established on-premises server infrastructure and configured Apache Airflow and Spark for distributed data processing
- Developed PL/SQL queries and ETL processes to extract and load data from Oracle ERP systems into data warehouse
- Implemented database optimization strategies and backup procedures for critical business data

Construction & Civil Engineering Leadership

- **Project Control Section Head** — Rowad Modern Engineering, Bridges Sector (Jan 2019 – Dec 2019)
- **Senior Project Control Engineer** — Rowad Modern Engineering, Head Office (Aug 2018 – Dec 2018)
- **Project Control Engineer** — Rowad Modern Engineering (Mar 2018 – Jul 2018)
- **Planner** — Concrete Plus (Apr 2015 – Mar 2018) - Multi-site project baseline development
- **Planner, QS Engineer** — Concrete Plus (Jan 2015 – Mar 2015)
- **Technical Office, QS Engineer** — Concrete Plus (Aug 2014 – Dec 2014)
- **Site Engineer** — Concrete Plus, Porto October Project (May 2014 – Jul 2014)
- **Site Engineer Training** — Orascom Construction Industries (Mar 2014 – May 2014)

EDUCATION

Cairo University — Bachelor of Science in Civil Engineering — May 2013

Overall Grade: Good (B) — *Project Grade:* Excellent (A)

Graduation Project: Steel Bridges Design

Orouba Language School — High School Diploma — 98.9% (405.5/410)

ADVANCED TECHNICAL SKILLS

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, Transformers (Hugging Face, ONNX, Transformers.js), XGBoost, LangChain, LlamaIndex, PEFT, RLHF/PPO/DPO, RAG, SFTTrainer, Lorax, SkyPilot, Llamafire, Computer Vision (CNNs, YOLO), NLP (BERT, GPT, T5, BLEU), Reinforcement Learning, Recommendation Systems, Advanced Clustering

Cloud & Infrastructure: AWS (SageMaker, Bedrock, Lambda, S3, Redshift), Azure (ML Studio, Synapse, OpenAI), GCP (Vertex AI, TPUs, Model Garden, BigQuery), Databricks (Unity Catalog, Delta Live Tables), Kubernetes, Docker, Terraform, CUDA, PyCUDA

Data Engineering & Analytics: Apache Spark, Hadoop, Airflow, Kafka, Snowflake, Delta Lake, Statistical Methods (ANOVA, t-tests, Regression Analysis), Model Evaluation (Cross-validation, KFold), Feature Engineering, Dimensionality Reduction

Development & Operations: Python, SQL, R, Java, C++, PostgreSQL, MongoDB, Vector Databases, MLflow, Jenkins, GitLab CI, pytest, IBM Watson OpenScale, Ansible, MPI4Py

LEADERSHIP & PROJECT MANAGEMENT

Methodologies: Agile, Scrum, Lean, Kanban, SAFe

Tools: Jira, GitLab, Taiga, OpenProject, MS Project, Primavera P6

Expertise: Team Leadership, Stakeholder Management, Risk Assessment, Strategic Planning, Cross-functional Collaboration

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Advanced AI & ML Specializations:

- **Deep Learning Professional Certificate** (IBM) - 5 courses
- **Natural Language Processing** (DeepLearning.AI) - 4 courses
- **RAG for Generative AI Applications** (IBM) - 4 courses
- **AI Developer Professional Certificate** (IBM)
- **AI Engineering Professional Certificate** (IBM)
- **LLMOps** (Duke University) - 5 courses
- **Data Science & Analytics:**
- **Applied Data Science** (IBM) - 5 courses
- **Data Analysis & Visualization Foundations** (IBM)
- **Introduction to Data Science** (IBM)
- **AI Foundations for Business** (IBM)

Cloud AI & Development:

- **Google Prompting Essentials** (Google) - 4 courses
- **Modern Project Leadership** (Coursera)
- **ChatGPT API for Developers**
- **Professional & Technical:**
- **Certified Data Scientist Professional** (Epsilon AI)
- **Project Management Professional (PMP)**
- **CCNA & CCNP Networking**
- **Python Programming Advanced**
- **Primavera P6 Project Management**
- **Steel Bridges Design Specialist**

Total: 12+ Specializations — 45+ Certificates

PERSONAL PROJECTS & ACHIEVEMENTS

- **Creative Leadership:** Guitarist in Iron Maiden tribute band, demonstrating artistic creativity and team collaboration
- **Digital Content Creation:** Built YouTube channel to 100,000+ subscribers, showcasing content strategy and audience engagement
- **Community Building:** Co-founded social media group ESS, developing community management and digital marketing skills
- **Media & Communication:** Co-hosted political podcast, enhancing public speaking and content creation abilities
- **Creative Writing:** Authored two science fiction novels, demonstrating storytelling and creative thinking capabilities
- **Community Service:** Active in street animal rescue operations in Maadi, showing commitment to social responsibility

LANGUAGES

Arabic: Native Proficiency — **English:** Professional Working Proficiency (Fluent)

References available upon request